

The Factorization Theorem

- How to find a sufficient statistic?
- **Factorization Theorem:** Let $X_1, \dots, X_n \sim f(x; \theta)$. The statistic $Y_1 = u_1(X_1, \dots, X_n)$ is a sufficient statistic for θ iff we can find two nonnegative functions, k_1 and k_2 , such that

$$f(x_1; \theta)f(x_2; \theta) \cdots f(x_n; \theta) = k_1[u_1(x_1, \dots, x_n); \theta]k_2(x_1, \dots, x_n),$$

where k_2 does not depend on θ .

- Let $X_1, \dots, X_n \sim N(\theta, \sigma^2)$, where $-\infty < \theta < \infty$ is unknown but $\sigma^2 > 0$ is known. If $\bar{x} = \sum_1^n x_i/n$, then the joint pdf is

$$\exp[-n(\bar{x} - \theta)^2] \left\{ \frac{\exp[-\sum_{i=1}^n (x_i - \bar{x})^2/2\sigma^2]}{(\sigma\sqrt{2\pi})^n} \right\}$$

Because the first factor depends on $x_1, \dots, x_n$ only through $\bar{x}$ and the second factor does not depend on θ , $\bar{X}$ is the sufficient statistic.

- Let $X_1, \dots, X_n$ denote a random sample from pdf

$$f(x; \theta) = \theta x^{\theta-1}, \quad 0 < x < 1, \quad \theta > 0.$$

The joint pdf is

$$\theta^n (\prod_{i=1}^n x_i)^{\theta-1} = [\theta^n (\prod_{i=1}^n x_i)^\theta] \frac{1}{\prod_{i=1}^n x_i}.$$

So, $u_1(X_1, \dots, X_n) = \prod_{i=1}^n X_i$ is the sufficient statistic for θ .

- **Pay attention to the support of the the distribution.**
- We have showed that $Y_1 = \min\{X_i\}$ is the sufficient statistic for θ when the pdf is $f(x; \theta) = e^{-(x-\theta)} I_{(\theta, \infty)}(x)$. Without considering the domain of the function, take $n=3$ and note that

$$e^{-(x_1-\theta)} e^{-(x_2-\theta)} e^{-(x_3-\theta)} = [e^{-\max x_i + 3\theta}] [e^{-x_1 - x_2 + x_3 + 3 \max x_i}].$$

You may think that $Y_3 = \max X_i$ is the sufficient statistic for θ . This is **WRONG**.

- $X_1, X_2, X_3 \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$. Show that $Y_1 = \frac{1}{6}(X_1 + 2X_2 + 3X_3)$ is not a sufficient statistic.

- $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Unif}([0, \theta])$. Find the sufficient statistic.